



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



NASA Core Flight System Health and Safety Application DoS Daily Aerospace Review

Threat Report

Classification	TLP:CLEAR
Published	2026-08-02
Version	1.0
Author	Lost Boys Cyber
Report Type	Threat Report

TLP:CLEAR | Lost Boys Cyber | NASA Core Flight System Health and Safety Application DoS Daily Aerospace Review - Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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1. Executive Summary

Flight-software frameworks and health/safety applications deserve sector attention because availability, command validation, telemetry integrity, and fault handling are safety-adjacent even when an issue is framed as denial-of-service rather than destructive compromise. Operators should prioritize asset ownership, build provenance, simulation testing, and mission-specific compensating controls.

Daily defender priority: convert the article/advisory signal into inventory, telemetry, and response checks instead of waiting for perfect attribution. This report is written for operational review and should be validated against local exposure.

2. Intelligence Requirements

Question	Current Answer	Confidence
What is the daily topic?	NASA cFS / flight-software denial-of-service exposure and safety-adjacent software assurance	Medium
Who is most exposed?	Spacecraft developers, satellite operators, aerospace integrators, mission-control software teams, and defense/government space programs.	Medium
Are hard IOCs available?	No reliable atomic IOCs were present in the selected daily sources; hunt logic is behavior-based.	High

3. Source Review & Confidence

Source	Contribution	Confidence
NASA Core Flight System project	Primary cFS project and software context	Medium
NASA IV&V cybersecurity and software assurance	Spaceflight software assurance context	Medium
CISA ICS advisories search	Government advisory source for industrial/embedded operational software	Medium

Where the selected sources are sector guidance or article indexes rather than primary incident advisories, confidence is higher for operational relevance than for attacker attribution or victim-specific details.

4. Threat Activity Overview

Flight-software frameworks and health/safety applications deserve sector attention because availability, command validation, telemetry integrity, and fault handling are safety-adjacent even when an issue is framed as denial-of-service rather than destructive compromise. Operators should prioritize asset ownership, build provenance, simulation testing, and mission-specific compensating controls.

The practical defender question is whether local controls can detect the likely exposure path and whether incident-response teams have playbooks for the business process affected by the threat.

5. Affected Sectors and Exposure

Exposure Pattern	Why It Matters
Public-facing service or portal	Creates scanning, credential abuse, or outage exposure.
Third-party / supplier dependency	Creates blind spots in logging, contractual response, and notification timelines.
Identity-federated SaaS or cloud platform	Enables data theft and persistence without traditional endpoint malware.
Operational process dependency	For aviation, healthcare, CI/CD, and software delivery, cyber events rapidly become business-impact events.

6. Aviation and Aerospace Sector Impact

Sector Element	Assessment
Affected aviation/aerospace segment	Spacecraft developers, satellite operators, aerospace integrators, mission-control software teams, and defense/government space programs.
Operational relevance	Flight-software frameworks and health/safety applications deserve sector attention because availability, command validation, telemetry integrity, and fault handling are safety-adjacent even when an issue is framed as denial-of-service rather than destructive compromise. Operators should prioritize asset ownership, build provenance, simulation testing, and mission-specific compensating controls.
Likely exposure paths	Third-party service providers, internet-exposed portals, remote access, supplier software, cloud mission platforms, and identity federation depending on the segment.
Supply-chain considerations	Validate managed service providers, airport/airline common-use systems, aerospace software suppliers, and data exchanges that can propagate disruption beyond one organization.
Defender actions for aviation/aerospace operators	Map critical business services, test degraded-operation procedures, preserve logs for common-use/mission systems, and pre-stage communications for regulator/customer notifications.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed or Plausible Technique	Evidence
Initial Access	T1199 / T1566	Exposure through supplier, public reporting, or social engineering
Execution	T1059 / T1219	Follow-on execution or remote-access activity where applicable
Impact	T1486 / T1499	Encryption, outage, data theft, or operational disruption

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
No confirmed atomic IOC set	Source limitation	Use behavioral detections and local telemetry correlation.
Supplier/cloud/remote-support access paths	Exposure class	Prioritize inventory and access review.
User-reported support contact or outage ticket	Triage signal	Correlate with authentication, endpoint, and network logs.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Unusual remote-support process execution or cloud-console activity	DeviceProcessEvents / Cloud audit logs	Detect initial access and hands-on-keyboard staging.
New data export, large download, or anomalous object access	SaaS / DLP / proxy logs	Detect collection and exfiltration.
Operational service outage or error burst following auth change	Application / service monitoring	Catch impact before customer-facing disruption widens.

9.2. Sigma / Detection Content

<pre> title: Suspicious Remote Support Or Supplier-Led Follow-On Activity status: experimental logsource: product: windows category: process_creation detection: remote_tools: Image contains: - 'QuickAssist' - 'AnyDesk' - 'TeamViewer' - 'ScreenConnect' suspicious_child: CommandLine contains: - 'whoami' - 'net user' - 'vssadmin' - 'rclone' condition: remote_tools or suspicious_child level: medium </pre>
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9.3. Hunt Query

<pre> DeviceProcessEvents where Timestamp > ago(14d) where FileName has_any ('QuickAssist.exe','AnyDesk.exe','TeamViewer.exe','ScreenConnect.ClientService.exe','powershell.exe','rclone.exe') summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Commands=make_set(ProcessCommandLine, 20) by DeviceName, AccountName, FileName order by LastSeen desc </pre>

10. Threat Hunting

10.1. Hunt Hypothesis

A supplier, social-engineering, cloud, or operational dependency path was abused before formal incident notification reached defenders.

10.2. Hunt Query

DeviceEvents where Timestamp > ago(14d) where ActionType has_any ('RemoteDesktopConnection', 'UserAccountCreated', 'AntivirusSignatureUpdateFailed', 'ExploitGuardNetworkProtectionBlocked') summarize Events=count(), Actions=make_set(ActionType, 20) by DeviceName, AccountName, bin(Timestamp, 1d) where Events > 5
CloudAppEvents where Timestamp > ago(14d) where ActionType has_any ('FileDownloaded', 'MassDownload', 'Add service principal', 'Consent to application', 'Update user') summarize Count=count(), Apps=make_set(Application, 20), IPs=make_set(IPAddress, 20) by AccountObjectId, AccountDisplayName where Count > 25

11. Risk Assessment

Dimension	Assessment
Likelihood	Medium; selected due to recency, exposure breadth, or aviation/aerospace operational relevance.
Impact	Medium to High depending on data sensitivity, operational dependency, and third-party access.
Confidence	Medium unless primary victim/advisory details are available.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC / IR	Add the included behavior hunts to daily review for exposed business units.
High	Identity / Cloud	Review third-party access, federation, consent grants, and anomalous downloads.
High	Operations	Validate degraded-service playbooks and contact paths with key suppliers.
Medium	Governance	Track source updates and revise the report if primary disclosures add IOCs or victim detail.

13. References

- [1] NASA Core Flight System project: <https://cfs.gsfc.nasa.gov/>
- [2] NASA IV&V cybersecurity and software assurance: <https://www.nasa.gov/centers-and-facilities/ivv/>
- [3] CISA ICS advisories search: https://www.cisa.gov/news-events/cybersecurity-advisories?f%5B0%5D=advisory_type%3A95